

Week 2 • Day 6

Model Evaluation Deep Dive

Ran cross-validation, optimised the classification threshold, and plotted the Precision-Recall curve

DATE	INTERN NAME	PROJECT	MENTOR
Day 13 of 28 Week 2		Customer Churn Prediction & LTV Engine	

■ Technologies Used Today

Python

Scikit-Learn

XGBoost

Pandas

Matplotlib

NumPy

joblib

■ Today's Key Metrics

5

CV Folds Run

~0.86

CV AUC Score

16

Thresholds Tested

1

Optimal Threshold Saved

■ Day Objectives

- Run 5-fold cross-validation for reliable AUC estimate
- Find optimal classification threshold using F1 score
- Plot Precision-Recall curve and calculate Average Precision
- Save optimal_threshold.pkl for use in the API
- Document final model performance metrics

■ Tasks Completed Today

1. Created src/models/model_evaluation.py
2. Loaded production_model.pkl and ran cross_val_score with cv=5, scoring='roc_auc'

3. Printed: 5-Fold CV AUC = mean $\pm$ std deviation
4. Generated predicted probabilities on test set
5. Tested 16 threshold values from 0.1 to 0.85 in 0.05 steps
6. Calculated F1-score at each threshold
7. Found optimal threshold (maximises F1) — typically around 0.35-0.45 for this dataset
8. Saved models/optimal_threshold.pkl using joblib
9. Calculated Precision-Recall curve and Average Precision score
10. Created 2-panel figure: PR curve (left) + Threshold vs F1 (right)
11. Saved reports/figures/13_threshold_optimisation.png
12. Committed all files to GitHub

■ Concepts Learned

- K-Fold Cross-Validation: trains and evaluates K times, averages results for stability
- Why a single train/test split gives unreliable estimates — depends on which rows happen to be in test
- Default threshold of 0.5 is rarely optimal for imbalanced datasets
- Lower threshold = higher recall (catch more churners) but lower precision (more false alarms)
- Average Precision (AP): area under PR curve, better than AUC for very imbalanced data
- The business decision: what cost ratio between false negatives and false positives should drive threshold choice?

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Scikit-Learn cross_val_score	1.3.2	✓ 5-fold CV
Scikit-Learn precision_recall_curve	1.3.2	✓ PR curve

■ Outputs / Deliverables Produced

- ◆ src/models/model_evaluation.py — cross-validation and threshold script
- ◆ models/optimal_threshold.pkl — saved best classification threshold
- ◆ reports/figures/13_threshold_optimisation.png — PR curve + F1 vs threshold
- ◆ 5-Fold CV AUC: mean $\pm$ std printed
- ◆ Optimal threshold identified and documented
- ◆ GitHub commit with evaluation script and threshold file

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
cross_val_score was very slow — took several minutes	Added n_jobs=-1 to cross_val_score to parallelise across all CPU cores

■ GitHub Commit Record

Commit Message	Week2-Day6: Cross-validation, threshold optimisation, PR curve — optimal threshold saved
Branch	main
Files Changed	src/models/model_evaluation.py, models/optimal_threshold.pkl, reports/figures/13_threshold_optimisation.png
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ 5-Fold CV AUC computed	■ CV std deviation < 0.02
■ 16 thresholds tested	■ Optimal threshold found
■ optimal_threshold.pkl saved	■ PR curve plotted
■ Average Precision calculated	■ Chart 13 saved
■ Threshold documented	■ Day 13 commit on GitHub

■ Self Assessment

8/10 Self Rating	5 Hours Spent	Week 2, Day 6 Progress	13/28 Day in Program
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■ Key Learnings & Reflections

- ★ Cross-validation AUC is more trustworthy than single-split AUC — always prefer it
- ★ Optimal threshold is usually below 0.5 for imbalanced datasets — default 0.5 misses too many churners
- ★ The PR curve is more informative than ROC when positive class is rare
- ★ Save the threshold alongside the model — they must be used together in the API

■ Plan for Tomorrow

- Implement SHAP explainability for the XGBoost production model
- Create SHAP summary plot, bar plot, and waterfall plot
- Explain a specific high-risk customer prediction with SHAP
- Save shap_explainer.pkl

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED